

- 16** Which of the following is a necessary condition for clear and stable interference patterns to be formed by waves from two sources?
- A** The sources of the waves are in phase.
 - B** The sources produce waves that are of similar amplitudes.
 - C** The sources produce waves that are propagating in the same direction.
 - D** The sources produce waves that are polarised in the same plane.

Ans: B

The sources need not be in phase, but their phase difference must be constant (coherence).

If the waves are of very different amplitudes, the contrast between the maxima and the minima will not be significant, which does not produce a clear interference pattern.

The paths of the waves must intersect, and not run parallel in the same direction.

The waves must not be polarised in perpendicular planes, but they can be unpolarised.